

Rockborne

Classification

Predicting categories from data

Objectives



What is Classification

The idea, and how it differs from regression.



Types of Classification

Algorithms and the kinds of classification problems.



Evaluating the Classifier

How we measure whether a model is any good.

Introduction to Classification



What is Classification

What is Classification?

- Classification predicts a class label for a given input example.
- Classes are also called targets, labels or categories.
- Used in credit approval, medical diagnosis, target marketing and fraud detection.

IMAGE PLACEHOLDER

Illustration: inputs sorted into labelled class buckets

Classification vs Regression

Both are supervised learning; the difference is the output variable.

Regression

- Predicts a numerical, continuous value.
- Example: predicting a student's exam percentage.
- Output is a quantity.

Classification

- Predicts a categorical, discrete label.
- Example: predicting pass or fail.
- Output is a class.

Common Terminology

Key concepts

- Classifier: the algorithm that maps input data to a category.
- Classification model: predicts the class or category for the data.
- Feature: an individual measurable property of what is observed.

The workflow

- Initialize: choose the classifier to use.
- Train: fit the model on features X and labels y .
- Predict: `predict(X)` returns the label y for new data.
- Evaluate: classification report, accuracy score, etc.

Introduction to Classification



Types of Classification

Classification Algorithms



Logistic Regression



Naive Bayes



K-Nearest Neighbours



Support Vector Machine



Decision Tree



Neural Networks

Eager vs Lazy Learners

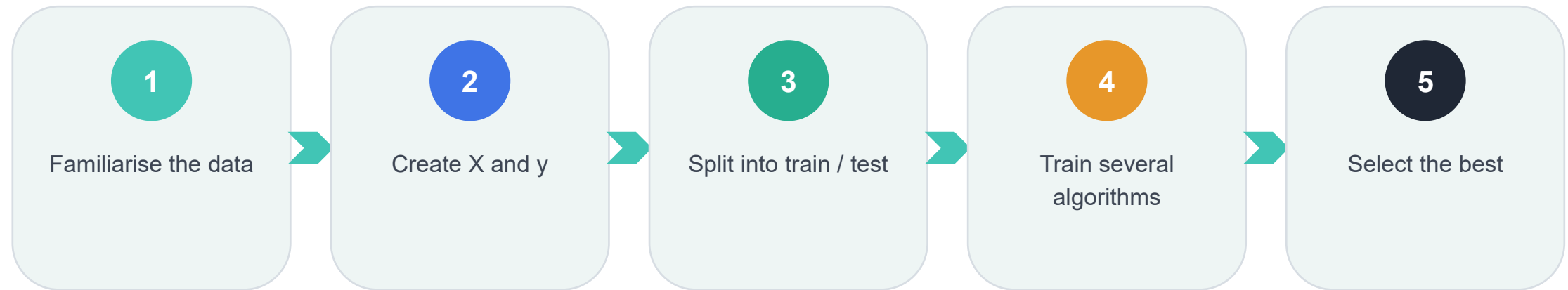
Eager learners

- Build the model from the training data before any new data arrives.
- Commit to a single hypothesis covering the whole space.
- More time to train, less time to predict.
- Decision Tree, Naive Bayes, Neural Networks.

Lazy learners

- Store the training data and wait until a test point appears.
- Classify using the most related stored data.
- Less time to train, more time to predict.
- K-Nearest Neighbours, Case-Based Reasoning.

Choosing the Best Algorithm



'No Free Lunch': every model carries some bias, so compare a few and pick the best for your data.

Types of Classification Problems



Binary

Two class labels (pass/fail, normal/abnormal). LogReg, KNN, Trees, Naive Bayes, SVM.



Multi-Class

More than two labels, one per example (face or species recognition). KNN, Trees, Random Forest, Gradient Boosting.



Multi-Label

Two or more labels per example (a photo with several objects). Multi-label Trees, Random Forest, Gradient Boosting.



Imbalanced

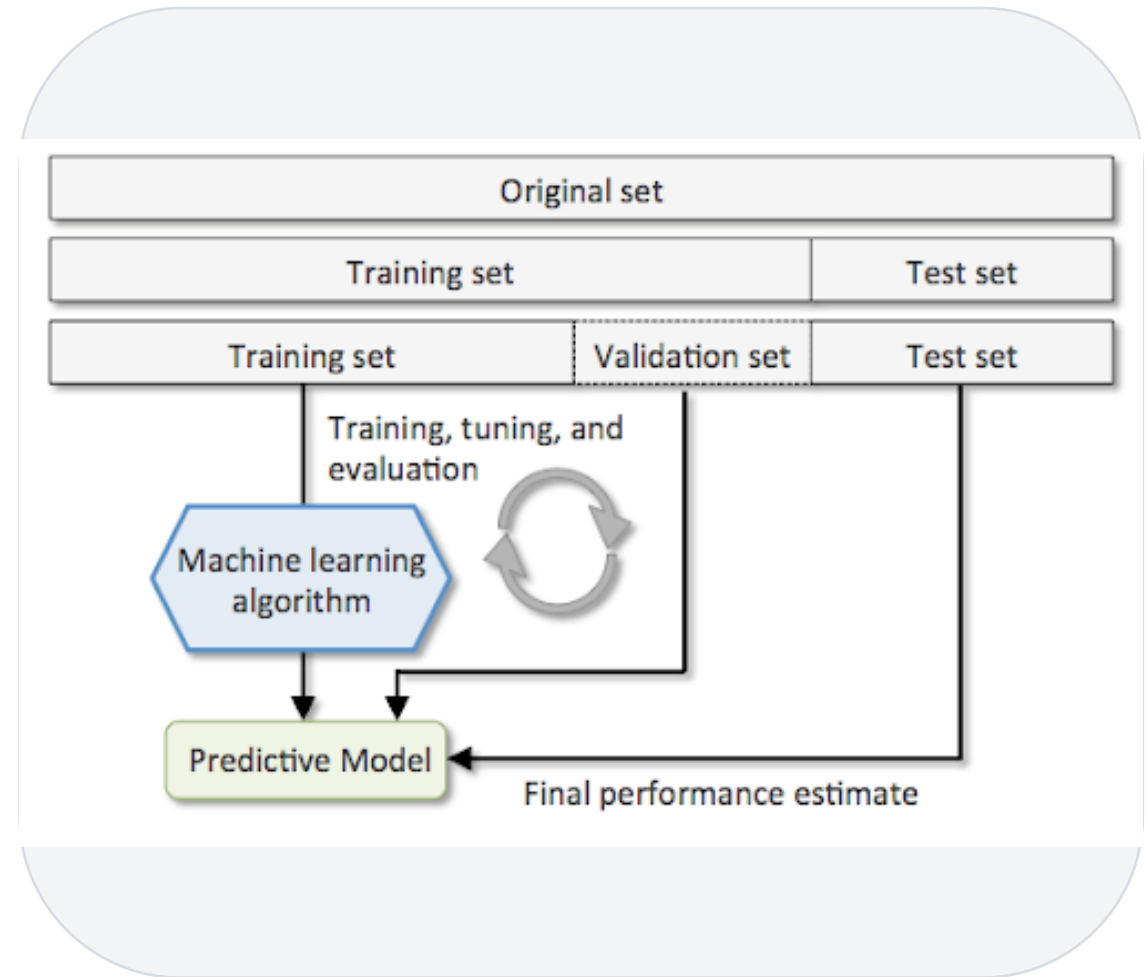
Classes unequally distributed (fraud, medical diagnosis). Resample plus specialised LogReg, Trees, SVM.

Introduction to Classification

Evaluating the Classifier

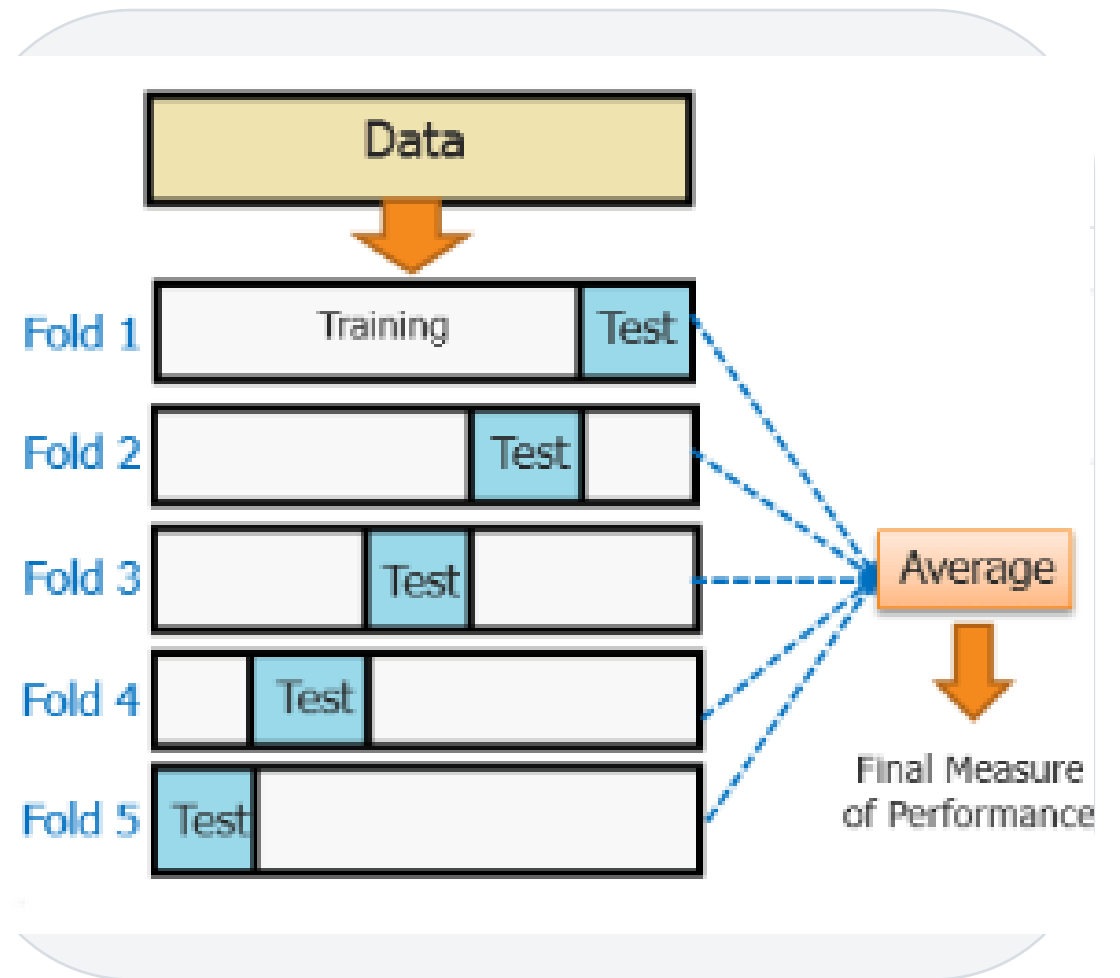
Holdout Method

- The common way to evaluate a classifier.
- Split into train and test sets (80/20 or 70/30).
- Measure accuracy on the held-out test set to judge fit.
- Simple but sensitive: a non-random split can bias results.



K-Fold Cross-Validation

- Partition the data into k equal, mutually exclusive folds.
- Hold one fold out for testing; train on the rest.
- Repeat until every fold has been the test set once.
- A stronger check against over-fitting than a single split.



The Classification Report

```
Run: svm x
C:\Users\Waseem\PycharmProjects\svm\venv\Scripts\python.exe C:/Users/Waseem/PycharmProjects/svm/svm.py
accuracy: 0.9649122807017544
precision: 0.9642857142857143
recall 0.9782608695652174

      precision    recall  f1-score   support

     0       0.97      0.94      0.96         90
     1       0.96      0.98      0.97        138

   accuracy          0.96         228
  macro avg       0.97      0.96      0.96         228
 weighted avg       0.96      0.96      0.96         228

Process finished with exit code 0
```

Accuracy: correct over all predictions.

Precision: relevant among retrieved.

Recall: relevant retrieved of all relevant.

F1: harmonic mean of precision and recall.

ROC Curve



1

Compares classification models visually.

2

Shows the true-positive vs false-positive trade-off.

3

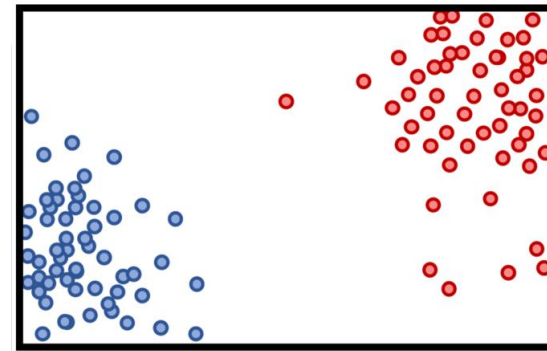
AUC (area under the curve) measures overall accuracy.

4

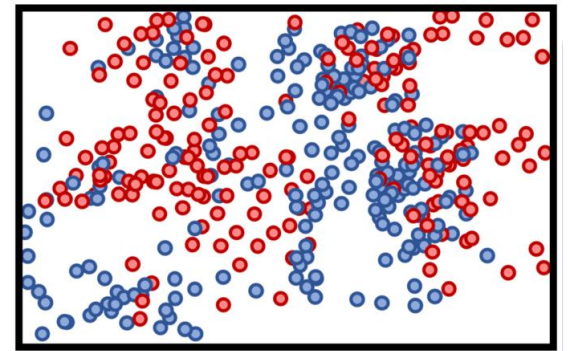
Closer to the diagonal is weaker; a perfect model has AUC = 1.

Entropy

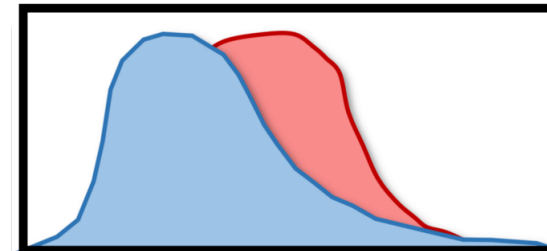
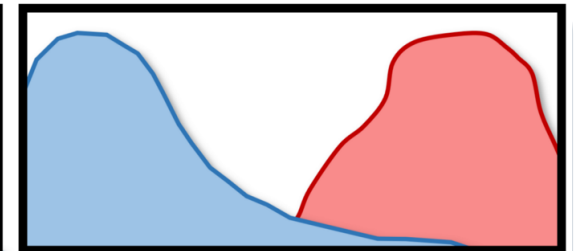
- Entropy measures uncertainty within a dataset.
- It appears everywhere in ML, from trees to neural networks.
- Information gain is the knowledge gained by adding a node.
- Information gain is inversely related to entropy.



Low Entropy



High Entropy

Low information gain
High entropyHigh information gain
Low entropy

Entropy Formulas

- The entropy of the target response (top).
- The entropy of the target given a feature X (bottom).
- The information-gain formula combines these entropy results.

Entropy Formula

WOMEN WHO
CODE X **CODE**
SILICON VALLEY DATA SCIENCE

$$E(y) = \sum_{i=(1,C)} -P(c_i) \times \log_2 P(c_i)$$

Entropy of
Target Response

Probability of
class c_i

$$E(y,X) = \sum_{c \in X} P(c_i) \times E(c_i)$$

Entropy of target class for
input feature $x_i \in X$

Probability of
class c of x_i

Entropy of
class c of x_i

Decision Tree - Step 2

WOMEN WHO
CODE X **CODE**
SILICON VALLEY DATA SCIENCE

Calculate the information gain of each input feature

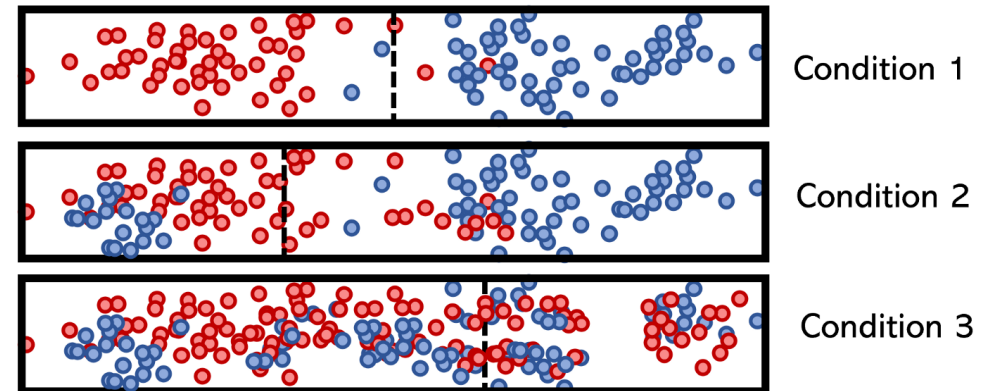
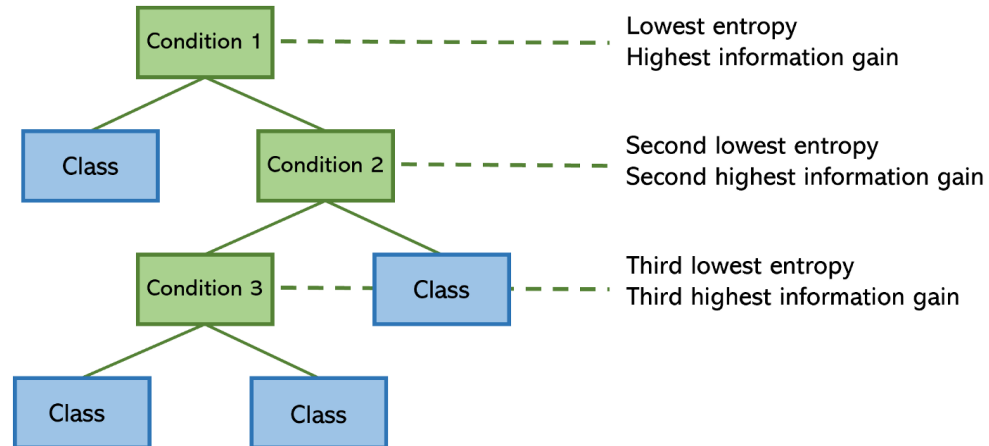
$$\text{Gain}(y, x_i) = E(y) \times E(y, x_i)$$

Information gain of
input feature $x_i \in X$

Entropy of
Target Response

Entropy of target class for
input feature $x_i \in X$

Decision Trees and Entropy



Trees compute feature entropy and arrange nodes to minimise it.

Information gain sets the order of attributes in the nodes.

Rockborne

Time to get hands-on in Python

We move from theory to practice with a guided walkthrough you run yourself in Python.



K-Nearest Neighbours - Introduction



KNN - Guided Walkthrough in Python

In this lesson, we explored



What is Classification

Predicting a class label; classification vs regression.



Types of Classification

Algorithms, eager vs lazy, and problem types.



Evaluating the Classifier

Holdout, k-fold, the report, ROC and entropy.

Digging Deeper?



Introduction to Classification Models

Video



Getting Started with Classification

GeeksforGeeks



Four types of predictive classification

Machine Learning Mastery

Questions?